

February 3, 2026

Editor
Psychological Methods
American Psychological Association

Dear Editor,

We are pleased to submit our manuscript, “A Tutorial in Dynamic Systems Simulation and Modelling With R, ctsem, and lme4: Couples’ Affect Dynamics Over Time,” for consideration for publication in *Psychological Methods*. This tutorial provides a comprehensive, hands-on introduction to dynamic systems modeling for psychological researchers, bridging the gap between theoretical concepts and practical implementation.

The manuscript introduces a structured workflow for building and understanding dynamic systems models using R and the ctsem package. We guide readers from familiar linear models in lme4 to more complex continuous-time models that can capture state-dependent change, system noise, covariate effects, interactions, and external inputs. Throughout the tutorial, we emphasize iterative model-building through simulations, model fitting, and visualizations, using affect dynamics in couples therapy as a running example.

We recognize that the manuscript may be longer than typical submissions. This length is partly due to the comprehensive nature of the tutorial, but also driven by the inclusion of many figures and code examples. We believe these elements are essential for readers to easily follow the tutorial and successfully implement the methods.

This tutorial addresses a growing need in psychological research, as interest in intensive longitudinal data and dynamic systems approaches continues to expand. By providing clear, step-by-step guidance with working code and visualizations, we aim to make appropriate use and understanding of these methods more accessible to the broader psychological research community.

Thank you for your time and consideration.

Charles C. Driver & Michael Aristodemou,
University of Zurich